Impact of State-Space Coordinate Representations on Recurrent Neural Network Performance in Chaotic Double Pendulum Prediction

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Abstract

Accurate prediction of chaotic physical systems using machine learning remains an open research challenge, particularly due to sensitivities arising from coordinate representation choices. This paper presents a systematic empirical study comparing three state-space coordinate encodings—Raw Angles, Trigonometric Components, and Cartesian Coordinates—as inputs to a shared Long Short-Term Memory (LSTM) recurrent neural network trained to predict the dynamics of a double pendulum. Three independent LSTM models are trained on identical datasets, each receiving the same physical information encoded differently. All models are evaluated in a consistent physical state space using Mean Squared Error and Tolerance-Based Accuracy. Results demonstrate that the Trigonometric representation achieves superior performance, while Raw Angle encoding suffers from boundary discontinuities and Cartesian encoding introduces rigid-link constraint violations under autoregressive rollout.

1 Introduction

The double pendulum is a canonical example of a deterministic nonlinear dynamical system exhibiting chaotic behaviour. Governed by coupled nonlinear ordinary differential equations (ODEs), the system's trajectories diverge exponentially from nearly identical initial conditions, rendering long-horizon prediction inherently difficult.

Machine learning surrogate models have been proposed as computationally efficient alternatives to numerical integration schemes such as Runge-Kutta 4th/5th order (RK45) [3, 2, 5]. However, a frequently overlooked design decision in such systems is the choice of **state representation**: the coordinate system used to encode the physical state as model input features. This choice has a direct impact on the stability of gradient-based training and the physical consistency of predictions.

A well-known challenge is the **angular wrap-around discontinuity** [6]. When an angular coordinate θ crosses the boundary $\pm\pi$, its scalar value undergoes a discontinuous jump despite representing a physically continuous motion. This discontinuity is problematic for neural networks, as it creates uninformative gradients in the loss landscape and disrupts coherence in sequential models.

In this paper, we investigate the impact of three coordinate representations on the prediction performance of a shared LSTM architecture [1] applied to the chaotic double pendulum. The study is motivated by the hypothesis that representations which preserve topological continuity of the angular state space will yield better-trained and more stable predictive models.

2 Physical System and Mathematical Formulation

2.1 System Description

The double pendulum consists of two rigid link rods of equal lengths $l_1 = l_2 = 1.0$ m and masses $m_1 = 5.0$ kg and $m_2 = 1.0$ kg, operating under gravitational acceleration $g = 9.81$ m/s^2. The system state at time t is defined by:

$$s_t = [\theta_1,\ \omega_1,\ \theta_2,\ \omega_2] \tag{1}$$

where θ_1, θ_2 are the angular displacements from the vertical, and $\omega_1 = \dot{\theta}_1$, $\omega_2 = \dot{\theta}_2$ are the corresponding angular velocities.

2.2 Equations of Motion

The equations of motion are derived using Lagrangian mechanics. The kinetic energy T and potential energy V of the system are:

$$T = \frac{1}{2}(m_1 + m_2)l_1^2\dot{\theta}_1^2 + \frac{1}{2}m_2 l_2^2\dot{\theta}_2^2 + m_2 l_1 l_2\,\dot{\theta}_1\dot{\theta}_2\cos(\theta_1 - \theta_2) \tag{2}$$

$$V = -(m_1 + m_2)\,g\,l_1\cos\theta_1 - m_2\,g\,l_2\cos\theta_2 \tag{3}$$

Applying the Euler-Lagrange equations to the Lagrangian $\mathcal{L} = T - V$:

$$\frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{\theta}_i}\right) - \frac{\partial \mathcal{L}}{\partial \theta_i} = 0, \qquad i \in \{1, 2\} \tag{4}$$

yields the coupled second-order nonlinear ODEs governing $\ddot{\theta}_1$ and $\ddot{\theta}_2$. These are derived symbolically using SymPy and integrated numerically via SciPy's RK45 solver to generate the training dataset.

3 Methodology

3.1 Dataset Generation

A dataset of 1,000 independent simulation rollouts of 25 time steps each ($dt = 0.02$ s) is generated. Initial conditions are drawn uniformly at random: $\theta_i \in [-\pi, \pi]$ rad and $\omega_i \in [-2, 2]$ rad/s. This yields approximately 24,000 one-step transition samples (s_t, s_{t+1}).

3.2 State Representations

The identical physical state is encoded into three distinct coordinate systems prior to model input:

3.2.1 Raw Angles

The state is provided directly as angular values:

$$S_{\text{raw}} = [\theta_1,\ \omega_1,\ \theta_2,\ \omega_2] \tag{5}$$

This encoding is susceptible to the wrap-around discontinuity at $\pm\pi$.

3.2.2 Trigonometric Representation

Each angle is decomposed into its sine and cosine components:

$$S_{\text{trig}} = [\sin\theta_1,\ \cos\theta_1,\ \omega_1,\ \sin\theta_2,\ \cos\theta_2,\ \omega_2] \tag{6}$$

This mapping is globally smooth and continuous. Angles are recovered from predictions via $\theta_i = \arctan2(\sin\theta_i, \cos\theta_i)$.

3.2.3 Cartesian Coordinates

The angular state is mapped to Cartesian mass positions and linear velocities:

$$x_1 = l_1 \sin\theta_1, \quad y_1 = -l_1 \cos\theta_1 \tag{7}$$

$$x_2 = x_1 + l_2 \sin\theta_2, \quad y_2 = y_1 - l_2 \cos\theta_2 \tag{8}$$

$$\dot{x}_1 = l_1\omega_1 \cos\theta_1, \quad \dot{y}_1 = l_1\omega_1 \sin\theta_1 \tag{9}$$

$$\dot{x}_2 = \dot{x}_1 + l_2\omega_2 \cos\theta_2, \quad \dot{y}_2 = \dot{y}_1 + l_2\omega_2 \sin\theta_2 \tag{10}$$

3.3 Model Architecture and Training

Three LSTM networks are trained, one per coordinate system, using identical hyperparameters to ensure a fair comparison. Each model receives a rolling window of the five most recent states as input and outputs the predicted next state.

Hyperparameter	**Value**
Model type	LSTM (2 layers)
Hidden units	128
History window	5 steps
Training epochs	60
Optimiser	Adam
Learning rate	0.001
Batch size	256
Train / Test split	80% / 20%

3.4 Evaluation Protocol

All models are evaluated in the physical state space $s = [\theta_1, \omega_1, \theta_2, \omega_2]$, with coordinate predictions mapped back to angles prior to comparison. Angular errors are computed using circular wrapping to correctly handle the $\pm\pi$ boundary.

Two metrics are reported:

- **Physical State MSE:** The Mean Squared Error between the predicted physical state $\hat{s}$ and the true physical state s across all N test samples:

$$\text{MSE} = \frac{1}{N}\sum_{i=1}^{N}(s_i - \hat{s}_i)^2 \tag{11}$$

- **Tolerance-Based Accuracy:** The percentage of test samples where the absolute error is within a small tolerance $\delta = 0.05$ across all four state variables. A prediction is considered accurate if:

$$|s_{\text{true}} - s_{\text{pred}}| \leq 0.05 \tag{12}$$

4 Results and Discussion

4.1 Single-Step Prediction Performance

Table 1: Comparative Evaluation of LSTM Models on Test Set

Representation	**Physical MSE**	**Accuracy ($\pm$0.05)**
Raw Angles	0.156554	3.71%
Trigonometric	0.025324	**23.71%**
Cartesian	**0.022907**	20.13%

4.2 Effect of Coordinate Topology

4.2.1 Raw Angles

The Raw Angles model achieves the lowest accuracy and diverges most rapidly during long-term rollout. The

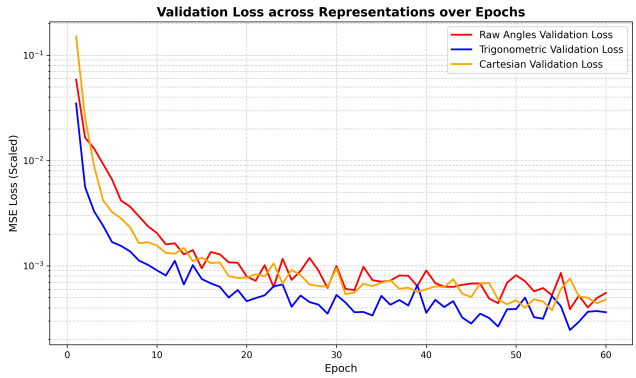


Figure 1: Validation loss scaling across training epochs for all three representations. The Trigonometric LSTM exhibits faster, more stable convergence due to the absence of boundary discontinuities.

discontinuous jump in angle values at $\pm\pi$ disrupts the LSTM's temporal learning, as adjacent states that are physically similar appear numerically distant at the boundary. This manifests as rapid accumulation of angular error in the autoregressive setting.

4.2.2 Trigonometric Representation

The Trigonometric model achieves the highest tolerance-based accuracy. Representing each angle as a sine-cosine pair maps the state space onto a smooth, compact topology with no discontinuities. Furthermore, angle reconstruction via arctan2 is exact, and the implicit unit-circle constraint ensures that the physical link lengths are preserved correctly throughout the prediction pipeline.

4.2.3 Cartesian Coordinates

Although Cartesian coordinates also avoid the wrap-around discontinuity and achieve marginally lower Mean Squared Error, they perform worse in terms of tolerance-based accuracy than the Trigonometric representation. Furthermore, the physical double pendulum is constrained to a rigid-link manifold defined by:

$$x_1^2 + y_1^2 = l_1^2 \tag{13}$$

$$(x_2 - x_1)^2 + (y_2 - y_1)^2 = l_2^2 \tag{14}$$

The LSTM has no mechanism to enforce these constraints on its outputs. As a result, small prediction errors cause the predicted positions to drift off this manifold, implicitly altering the effective link lengths. When angles are reconstructed from these unconstrained Cartesian coordinates, the constraint violation introduces additional error that compounds under autoregressive rollout, leading to earlier chaotic divergence compared to the Trigonometric model.

5 Conclusion

This study demonstrates that the choice of coordinate representation is a critical factor in the design of recurrent neural network surrogate models for chaotic dynamical systems. The following conclusions are drawn from the comparative evaluation:

- **Raw angle encoding** is the least suitable representation due to the boundary discontinuity at $\pm\pi$, which disrupts temporal coherence in LSTM training.
- **Trigonometric encoding** is the most suitable representation, providing a globally smooth, continuous state space that preserves physical link-length constraints implicitly.
- **Cartesian coordinate encoding** improves over raw angles but underperforms the trigonometric approach due to unconstrained manifold drift during prediction.

These findings suggest that future work on physics-based neural network surrogate modelling should carefully consider the topological properties of the chosen coordinate system. Promising directions include explicit constraint projection for Cartesian outputs and physics-informed loss functions that penalise violations of conservation laws [4].

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